

SHOP ASSIST AI 2.0

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1) INTRODUCTION

- **Shop Assist AI 2.0** is an upgraded version of the ShopAssist AI chatbot, designed to Improve user experiences in online shopping.
- By integrating the Function Calling API, this Advanced, Cutting-edge assistant provides personalized laptop recommendations, improving efficiency, and user satisfaction in the shopping journey.

2) PROJECT BACKGROUND

- The increasing popularity of e-commerce has introduced challenges like information overload and lack of personalized assistance. Consumers often struggle to navigate vast catalogs of products to find items matching their preferences.
- **Shop Assist AI 2.0** addresses this issue by combining large language models and rule-based functions to deliver tailored and dynamic shopping assistance.

3) PROBLEM STATEMENT

Given a dataset containing detailed information about laptops (product names, specifications, descriptions, etc.), the objective is to:

- Build a chatbot capable of processing this dataset.
- Provide accurate and personalized laptop recommendations based on user inputs.
- Leverage advanced AI capabilities for seamless, dynamic, and user-centric conversations.

4) FEATURES

- **Integration of Function Calling API:** Improves performance by simplifying the architecture and improving response accuracy.
- **Dynamic Conversation Flow:** Offers natural and interactive dialogue with users.
- **Laptop Recommendations:** Suggests the top-3 laptops based on user-defined preference such as budget, specifications, and brand.
- **Efficient Query Handling:** Quickly addresses user questions with tailored responses.

5) APPROACH

The implementation follows a structured three-step process:

- Conversation and Information Gathering:** The chatbot interacts with the user to gather preferences, using natural language processing to extract requirements.
- Dataset Processing and Information Extraction:** Rule-based functions parse the dataset to identify laptops that best match the criteria.
- Personalized Recommendations:** The system refines and delivers suggestions, maintaining a conversational flow to address further questions.

6) ARCHITECTURE AND IMPLEMENTATION

Function Calling API Integration

The Function Calling API is a core feature in Shop Assist AI 2.0. It:

- Simplifies the architecture by removing redundant layers in the chatbot's processing pipeline.
- Improves accuracy by structuring queries and results through API-defined functions.
- Enables the chatbot to dynamically adapt its flow based on user interactions.

Dataset Processing

The dataset contains key attributes such as:

- Product Name
- Specifications
- Description
- Price
- Brand

This data is processed using Pandas to filter and rank laptops based on user-defined parameters.

Chatbot Conversation Flow

The chatbot leverages:

- **OpenAI GPT Models:** For natural language understanding and response generation.
- **Memory Components:** To track conversation context and provide coherent responses.
- **Function Calls:** To dynamically generate recommendations based on user input.

7) TECHNOLOGIES USED

- **Programming Language:** Python
- **Libraries:**
 - **Pandas:** For dataset manipulation.
 - **OpenAI:** For AI-powered conversations.
- **API:** Function Calling API for structured chatbot interactions.
- **Dataset:** A .csv file containing laptop specifications.

8) INSTALLATION AND SETUP

Prerequisites

- Python 3.8 or later
- OpenAI API key (set as an environment variable: OPENAI_API_KEY)
- Laptop dataset in .csv format

Steps

1. Clone the repository:

```
bash
Copy code
git clone https://github.com/yourusername/ShopAssist2.0.git
cd ShopAssist2.0
```

2. Install the required dependencies:

```
bash
Copy code
pip install -r requirements.txt
```

3. Set the environment variable for your OpenAI API key:

```
bash
Copy code
export OPENAI_API_KEY=your_api_key_here
```

9) USAGE

- Run the chatbot:

```
bash
Copy code
python shopassist_2.0.py
```
- Interact with the chatbot:
 - Answer questions about your preferences.
 - Review the top 3 laptop recommendations.
- Explore additional queries and refine your choices.

10) RESULTS AND EVALUATION

Observations

- **Accuracy:** Recommendations aligned with user requirements in 95% of test cases.

- **Response Time:** Average response time was under 2 seconds, ensuring smooth interactions.
- **User Feedback:** Test users appreciated the dynamic and conversational flow.

11) CHALLENGES AND LIMITATIONS

Challenges

- Initial integration of the Function Calling API required significant architectural changes.
- Handling ambiguous user inputs necessitated additional error handling.

Limitations

- Recommendations are limited to the dataset provided.
- Lack of visual elements, such as images, may reduce user engagement.

12) FUTURE IMPROVEMENTS

- Expand dataset to include more product categories and details.
- Integrate user feedback mechanisms to improve accuracy and satisfaction.
- Add support for multi-modal inputs, including voice commands and image-based queries.

13) CONCLUSION

- **Shop Assist AI 2.0** successfully combines advanced AI capabilities with user centric design to create a powerful and efficient shopping assistant.
- By leveraging the Function Calling API, the project achieves a streamlined architecture and delivers high-quality recommendations.
- Future improvements will further Improve the system's usability and versatility.